

Advanced Full Stack Development with AI

13 Modules · 8 Course Combos · From HTML to a Deployed, AI-Tested Full Stack App

Syllabus Overview — 13 Modules

Module 1 — HTML Structure, forms, semantic HTML5, accessibility & responsive markup. • Document structure & semantics • Forms & input types • Tables, media, accessibility • Responsive HTML (picture, srcset)	Module 2 — CSS Styling, layout systems and modern responsive design. • Box model, positioning, selectors • Flexbox & CSS Grid • Transitions, transforms, animations • Responsive design, variables, dark mode
Module 3 — Bootstrap Rapid, responsive UI with a component-driven CSS framework. • Grid system & utilities • Navbar, cards, modals, carousel • Forms & tables • Mobile-first responsive patterns	Module 4 — JavaScript Core logic, DOM, and asynchronous programming. • Variables, functions, closures • DOM manipulation & events • Promises, async/await, fetch API • ES6+, JSON, error handling
Module 5 — TypeScript Type-safe JavaScript for larger, more reliable codebases. • Types, interfaces, generics • Classes & inheritance • Utility types • Typing DOM & React	Module 6 — React JS Component-driven single page applications. • Components, props, state, hooks • React Router & Context API • Fetching data, custom hooks • Performance & deployment
Module 7 — Next.js Production-grade React with SSR, routing & optimization. • File-based routing & layouts • Server vs client components • API routes & data fetching • Image/font optimization, deployment	Module 8 — Python & Advanced Python Programming fundamentals through OOP and beyond. • Data types, control flow, functions • OOP — classes, inheritance, polymorphism • File handling & exceptions • Iterators, generators, decorators
Module 9 — Django + REST Framework MVT web apps and REST APIs, connected to React/Next.js. • Models, ORM, admin panel • Forms, auth, sessions • DRF serializers & viewsets • JWT auth, CORS, connecting to frontend	Module 10 — Java (Core + OOP) Strongly-typed programming fundamentals & OOP. • Data types, arrays, strings • OOP — inheritance, polymorphism • Collections framework • Exception handling, generics, lambdas

Module 11 — Spring Boot Enterprise-grade REST APIs on the JVM. • REST controllers & layered architecture • Spring Data JPA • Validation & exception handling • Spring Security & JWT auth	Module 12 — Node.js + Express JavaScript on the backend — REST APIs & middleware. • Routing, middleware, controllers • Connecting MongoDB/SQL • Error handling & validation • JWT auth, connecting to frontend
Module 13 — SQL, PostgreSQL & Deployment Relational databases and shipping your app to production. • Joins, aggregates, subqueries, transactions • PostgreSQL — JSONB, indexes, window functions • Git/GitHub, environment variables • Deployment (Render/Vercel), Docker & CI/CD awareness	

8 Course Combos, Built From These Modules

- Frontend Combo 1 (React) — 7 wk
- Frontend Combo 2 (React + Next.js) — 10 wk
- React + Node/Express — 12 wk
- React + Django — 12 wk
- React + Spring Boot — 12 wk
- Next.js + Node/Express — 12 wk
- Next.js + Django — 12 wk
- Next.js + Spring Boot — 12 wk